Computer Networks

Module 1

Overview of the Internet
Protocol layering (Book 1 Ch 1)

Application Layer

- Application-Layer Paradigms
- Client-server applications
  - World Wide Web and HTTP
  - FTP
  - Electronic Mail
  - DNS
- Peer-to-peer paradigm
  - P2P Networks
- Case study: BitTorrent (Book 1 Ch 2)

1. **Introduction**
   - Providing Services
   - Application-Layer Paradigms
2. **Standard Client-Server Applications**
   - World Wide Web and HTTP
   - FTP
   - Electronic Mail
   - Domain Name System (DNS)
3. **Peer-to-Peer Paradigm**
   - P2P Networks

2.3.6 Domain Name System (DNS)

Definition

The Domain Name System (DNS) is a distributed client-server application that maps human-readable domain names to IP addresses.

- TCP/IP communication uses IP addresses.
- Humans prefer meaningful names instead of numeric addresses.
- DNS acts as the Internet's directory service.
- DNS translates domain names into IP addresses before communication begins.

Why Do We Need DNS?

- Every Internet host has a unique IP address.
- Remembering numeric addresses is difficult.
- Users prefer names such as
  - www.google.com
  - www.vjcet.org
  - www.forouzan.biz
- DNS converts these names into IP addresses automatically.

Key Idea

DNS is similar to a telephone directory that maps names to phone numbers.

Why is DNS Distributed?

A single centralized database is impractical because:

- Huge size of the Internet
- Heavy processing load
- Single point of failure
- Slow response for global users

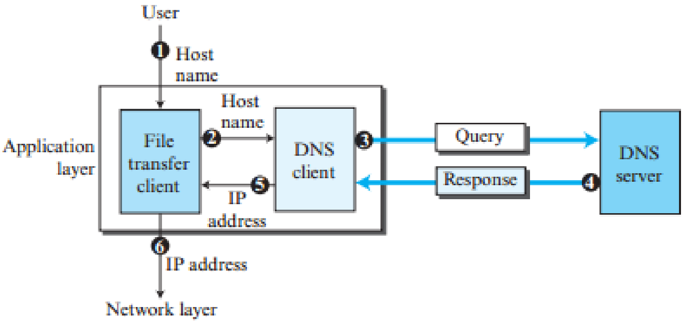
Solution

DNS distributes information across thousands of DNS servers worldwide.

DNS Client-Server Architecture

- Application needs an IP address.
- Application sends the hostname to the DNS client (Resolver).
- Resolver contacts a DNS server.
- DNS server returns the corresponding IP address.
- Application communicates using the returned IP address.

Figure 2.35 *Purpose of DNS*



DNS Name Resolution Steps

1. User enters the hostname.
2. File transfer client receives the hostname.
3. Client sends hostname to DNS resolver.
4. Resolver queries the DNS server.
5. DNS server returns the IP address.
6. Client communicates with the destination host.

Connections Required in DNS

Important Observation

DNS resolution itself requires communication before the actual application communication begins.

Two separate connections are required.

1. DNS Client $\rightarrow$ DNS Server
2. Application Client $\rightarrow$ Destination Server

Exam Point

DNS lookup always occurs before application communication.

Name Space

Definition

A Name Space is a collection of unique names assigned to Internet hosts.

DNS supports two types of name spaces.

- Flat Name Space
- Hierarchical Name Space

Flat Name Space vs Hierarchical Name Space

Flat Name Space	**Hierarchical Name Space**
Names have no structure	Names consist of multiple levels
Requires central management	Management is distributed
Suitable only for small systems	Suitable for the Internet
Difficult to maintain	Easy to expand
No hierarchy	Tree-based hierarchy

Conclusion

DNS uses a hierarchical name space.

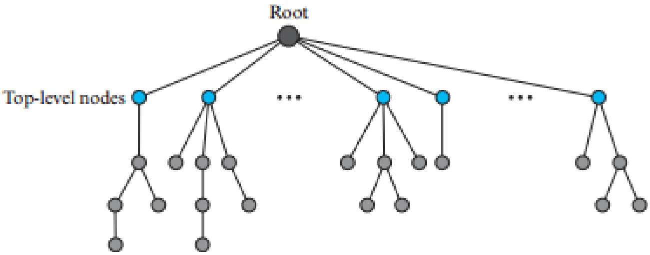
Domain Name Space

Definition

DNS organizes names using a hierarchical inverted-tree structure called the **Domain Name Space**.

- Root node is at the top.
- Maximum depth is 128 levels.
- Each node represents a domain.
- Every node has a unique name.

Figure 2.36 *Domain name space*



Labels in DNS

Label

Each node in the DNS tree has a label.

Properties

- Maximum length $= 63$ characters
- Root label is a null (empty) string
- Children of the same parent must have different labels
- Ensures uniqueness of domain names

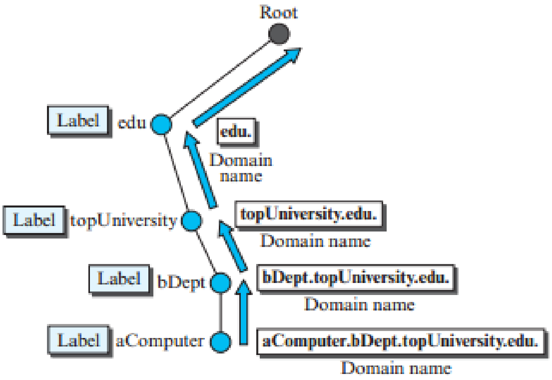
Domain Name

Definition

A domain name is a sequence of labels separated by dots (.).

- Read from the node toward the root.
- Root label is empty.
- Therefore every complete domain name ends with a dot (.).
- Example:
  - cs.vjcet.ac.in.
  - www.google.com.

Figure 2.37 *Domain names and labels*



FQDN vs PQDN

FQDN	**PQDN**
Fully Qualified Domain Name	Partially Qualified Domain Name
Ends with root dot	Does not end with root dot
Globally unique	Used within local domains
Complete path to root	Resolver adds missing suffix
Example: www.google.com.	www.google

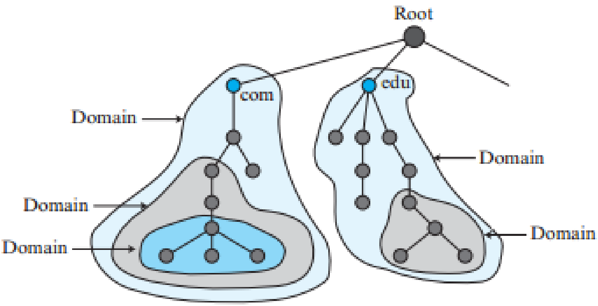
Domains and Subdomains

Domain

A domain is a subtree within the DNS hierarchy.

- Every node represents a domain.
- Domains can be divided into smaller subdomains.
- Delegation simplifies administration.
- Each organization manages its own subdomains.

Figure 2.38 *Domains*



Distribution of Name Space

Why distribute DNS?

- Huge amount of data
- Better scalability
- Improved reliability
- Faster response
- No single point of failure

Solution

Divide the DNS tree into multiple domains handled by different servers.

Hierarchy of Name Servers

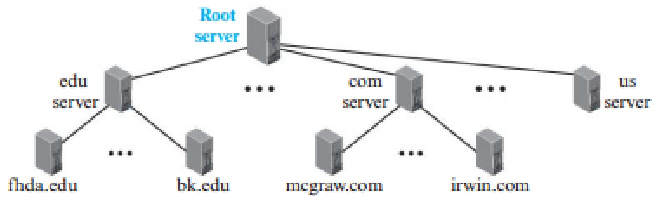
DNS information is distributed among multiple servers.

- Root Servers
- Top-Level Domain (TLD) Servers
- Authoritative Servers

Benefits

- Load sharing
- Fault tolerance
- Faster lookup

Figure 2.39 *Hierarchy of name servers*



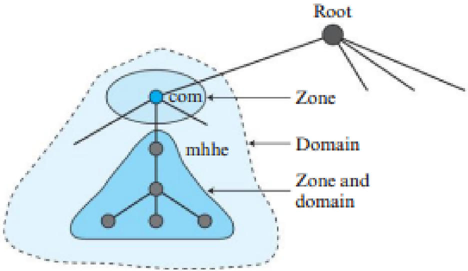
Zone

Definition

A Zone is the portion of the DNS tree managed by one DNS server.

- Server stores information for its zone.
- Zone may be equal to an entire domain.
- Large domains are divided into multiple zones.
- Lower-level servers manage delegated zones.

Figure 2.40 *Zone*



Root Server

Root Server

A root server represents the top of the DNS hierarchy.

Characteristics

- Knows every Top-Level Domain.
- Does not store every hostname.
- Delegates requests to TLD servers.
- Multiple root servers exist worldwide.

Primary Server vs Secondary Server

Primary Server	**Secondary Server**
Creates zone file	Copies zone file
Updates records	Cannot modify records
Stores master copy	Stores replicated copy
Source of updates	Obtains updates from primary
Maintains database	Provides redundancy

Exam Point

Both Primary and Secondary servers are authoritative.

DNS in the Internet

Role of DNS

DNS helps Internet applications translate host names into IP addresses before communication begins.

- Used by Web browsers
- Used by FTP clients
- Used by E-mail systems
- Used by almost every Internet application

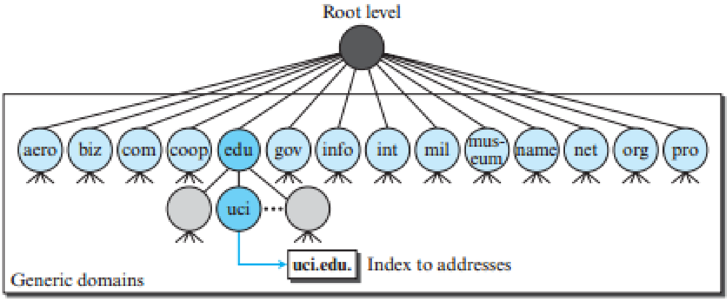
Generic Domains

Definition

Generic domains classify organizations based on their purpose or function.

- Top-level domains (TLDs)
- Managed by DNS authorities
- Easy identification of organization type

Figure 2.41 *Generic domains*



Common Generic Domain Labels

Label	Description
com	Commercial organizations
edu	Educational institutions
gov	Government institutions
org	Non-profit organizations
net	Network support centers
mil	Military organizations
int	International organizations
biz	Business organizations
info	Information providers
name	Personal names
museum	Museums
coop	Cooperative organizations
pro	Professional organizations
aero	Airlines and aerospace

Country Domains

Definition

Country domains use two-letter country codes.

Examples

- in – India
- us – United States
- uk – United Kingdom
- jp – Japan
- au – Australia

Second-level domains may further classify organizations.

Country Domain Example

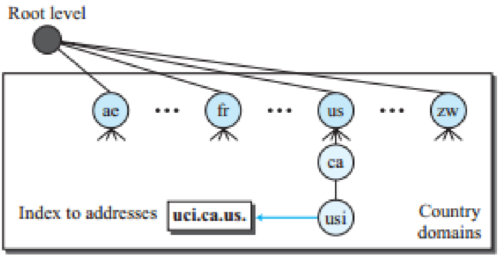
Example:

$$uci.ca.us.$$

Meaning

- uci = University of California Irvine
- ca = California
- us = United States

Figure 2.42 *Country domains*



Name Resolution

Definition

Mapping a domain name to an IP address is called Name Resolution.

Components

- Resolver (DNS Client)
- Local DNS Server
- Root Server
- TLD Server
- Authoritative DNS Server

Recursive Resolution

Definition

In recursive resolution, every DNS server contacted is responsible for obtaining the final answer.

Characteristics

- Client sends one request.
- Local DNS performs all remaining lookups.
- Final IP address returned to client.

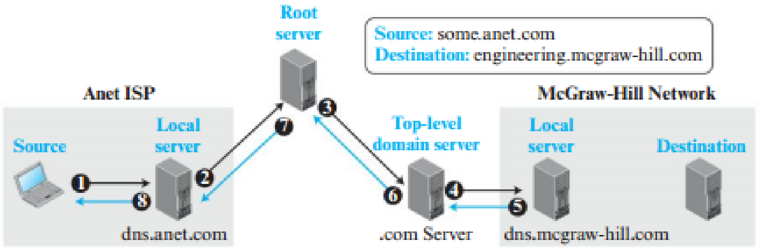
Recursive Resolution Process

Steps

1. Resolver sends request to local DNS server.
2. Local DNS contacts Root Server.
3. Root contacts Top-Level Domain Server.
4. TLD contacts Authoritative Server.
5. Authoritative Server returns IP address.
6. Response travels back through all servers.
7. Local DNS caches the result.
8. Resolver receives the final IP address.

Recursive Resolution Example

Figure 2.43 *Recursive resolution*



Observation

Every server participates until the final answer reaches the client.

Iterative Resolution

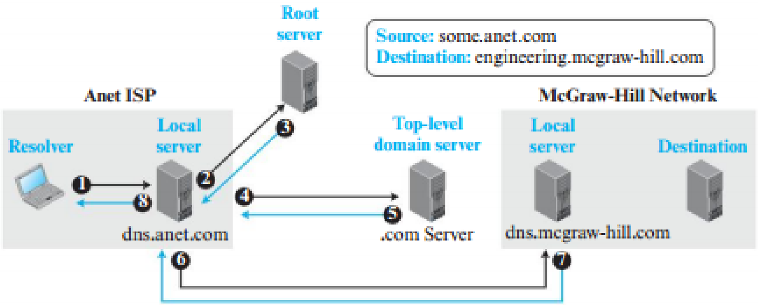
Definition

In iterative resolution, each DNS server returns the address of the next server instead of obtaining the answer itself.

Characteristics

- Server gives referral.
- Client (or local DNS) performs the next lookup.
- Fewer responsibilities on intermediate servers.

Figure 2.44 *Iterative resolution*



Recursive vs Iterative Resolution

Recursive Resolution	Iterative Resolution
Server finds the final answer	Server returns next DNS server address
Less work for client	More work for client/local resolver
More processing at DNS servers	Less processing at DNS servers
Single request from client	Multiple requests may be required
Common between resolver and local DNS	Common between DNS servers

DNS Caching

Definition

DNS caching stores recently resolved mappings so future requests can be answered faster.

- Improves lookup speed
- Reduces network traffic
- Reduces load on authoritative servers
- Frequently accessed records are reused

Time To Live (TTL)

TTL

TTL specifies how long a cached DNS record remains valid.

- Specified by authoritative server
- Measured in seconds
- Cache is deleted after TTL expires
- Fresh query is generated after expiration

Exam Point

TTL prevents outdated DNS information from being used.

Resource Records (RR)

Definition

DNS stores information using Resource Records (RRs).

Each Resource Record contains:

- Domain Name
- Type
- Class
- TTL
- Value

Resource Record Format

General Format

(Domain Name, Type, Class, TTL, Value)

Meaning of each field:

- Domain Name – Host name
- Type – Resource type
- Class – Network type (usually IN)
- TTL – Cache lifetime
- Value – Actual data

Common DNS Resource Record Types

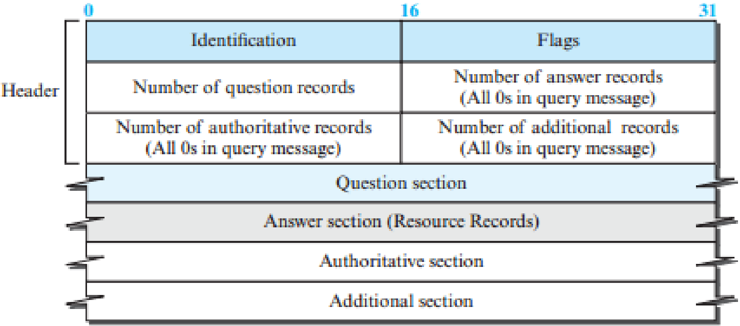
Type	Purpose
A	IPv4 Address
AAAA	IPv6 Address
NS	Authoritative Name Server
MX	Mail Server
CNAME	Alias Name
SOA	Start of Authority

DNS Message Format

Every DNS message contains:

- Header
- Question Section
- Answer Section
- Authority Section
- Additional Information Section

Figure 2.45 *DNS message*



Note:
The query message contains only the question section.
The response message includes the question section,
the answer section, and possibly two other sections.

DNS Header Fields

Header contains

- Identification
- Flags
- Number of Questions
- Number of Answers
- Authority Record Count
- Additional Record Count

Purpose

- Matches responses with queries
- Indicates errors
- Identifies message type

DNS Query and Response

Query Message	Response Message
Contains Question Section	Contains Question + Answer Sections
Sent by Resolver	Sent by DNS Server
Requests Mapping	Returns Mapping
May contain one or more questions	May contain authority and additional records

Example: Using nslookup

Example command:

UNIX / Windows

```
nslookup www.forouzan.biz
```

Output

- Domain Name
- Corresponding IP Address

Exam Point

The `nslookup` utility is used to retrieve DNS name-to-address mappings.

DNS Encapsulation

DNS can use both UDP and TCP transport protocols.

Well-known Port

DNS Server Port = **53**

- UDP is the default transport protocol.
- TCP is used when reliable transfer is required.
- Choice depends on response size and application.

UDP vs TCP in DNS

UDP	TCP
Default protocol	Used for large responses
No connection establishment	Connection-oriented
Fast	Reliable
Response size $<$ 512 bytes	Response size $>$ 512 bytes
Normal DNS queries	Zone transfer and large responses

Exam Point

If a UDP response exceeds 512 bytes, the server sets the TC (Truncated) bit and the resolver repeats the request using TCP.

Registrars

Definition

A Registrar is a commercial organization accredited by ICANN to register domain names.

Responsibilities

- Verifies domain name uniqueness
- Registers new domains
- Updates DNS database
- Charges registration fees

Dynamic DNS (DDNS)

Definition

Dynamic Domain Name System (DDNS) automatically updates DNS records whenever IP addresses change.

Why DDNS?

- Hosts frequently change IP addresses.
- Manual updates are impractical.
- DHCP can automatically notify DNS servers.

DDNS Update Process

1. Host receives a new IP address.
2. DHCP informs the Primary DNS Server.
3. Primary Server updates the Zone File.
4. Secondary Servers receive updates.
5. DNS database becomes synchronized.

Result

Automatic DNS maintenance without manual intervention.

Active vs Passive Notification

Active Notification	**Passive Notification**
Primary server informs secondary server	Secondary server periodically checks for updates
Immediate synchronization	Synchronization after polling interval
Faster updates	Slightly delayed updates

Zone Transfer

After notification, the Secondary Server performs a Zone Transfer.

DNS Security Threats

DNS is vulnerable to several attacks.

- Eavesdropping
- DNS Spoofing
- Message Modification
- Fake DNS Responses
- Denial-of-Service (DoS)

Impact

Users may be redirected to malicious websites or lose access to services.

DNSSEC (DNS Security Extensions)

Purpose

DNSSEC enhances DNS security by providing authentication and integrity.

Provides

- Message Origin Authentication
- Data Integrity
- Protection against forged DNS responses

Does NOT provide

- Confidentiality
- Complete DoS protection

Key Takeaways

- DNS translates domain names into IP addresses.
- DNS follows a hierarchical distributed architecture.
- Domains are divided into Zones managed by Name Servers.
- DNS supports Recursive and Iterative Resolution.
- Resource Records store DNS information.
- DNS mainly uses UDP Port 53.
- TCP is used for Zone Transfer and large responses.
- DDNS automates DNS updates.
- DNSSEC improves DNS security.

Module Summary

Topics Covered

- Introduction to DNS
- Name Space
- Domain Name Space
- Domains and Zones
- Name Servers
- Generic and Country Domains
- Recursive and Iterative Resolution
- DNS Caching
- Resource Records
- DNS Messages
- UDP vs TCP
- DDNS
- DNS Security and DNSSEC